DATA SCIENCE MINOR PROJECT
PROJECT REPORT

(Project Semester January-April 2025)

Road Crash Data Analysis & Visualization for South Australia (2019–2023)

Submitted by

Manish

Registration No: 12304562

Programme and Section: B. tech. CSE, K23SH

Course Code: INT217

Under the Guidance of

Mrs. Aashima Bansal (UID: 28968)

Discipline of CSE/IT

Lovely School of Computer Science and Engineering

Lovely Professional University, Phagwara

CERTIFICATE

This is to certify that Manish bearing Registration no. 12304562 has completed INT217 project titled, “Road Crash Data Analysis & Visualization for South Australia (2019–2023)” under my guidance and supervision. To the best of my knowledge, the present work is the result of his original development, effort and study.

Signature and Name of the Supervisor:

Designation of the Supervisor:

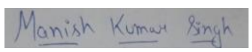
School of Computer Science and Engineering
Lovely Professional University
Phagwara, Punjab.

Date: 10-04-2025

DECLARATION

I, Manish Kumar Singh, student of B. Tech. under CSE/IT Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 10-04-2025

Signature: 

Registration No. 12304562

Name of the student: Manish Kumar Singh

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my mentor and guide, Mrs. Aashima Bansal , for their continuous support, motivation, and insightful suggestions throughout this project. Their expertise and guidance were instrumental in shaping the direction of this research.

I would also like to extend my thanks to the faculty members of the School of Computer Science and Engineering at Lovely Professional University for providing valuable feedback and resources. Special thanks to the university administration for offering access to necessary software tools and datasets that made this project possible.

Lastly, I am grateful to my peers and family for their encouragement and support during this endeavor.

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1. Introduction

Road accidents remain one of the leading causes of fatalities and injuries worldwide, with significant social and economic repercussions. According to the World Health Organization (WHO), approximately 1.3 million people die each year due to road traffic crashes, with millions more sustaining serious injuries. South Australia, like many regions, faces challenges in mitigating road accidents, necessitating data-driven approaches to understand and address the underlying causes.

This project focuses on analyzing road crash data from South Australia between 2019 and 2023. The primary objectives are to:

Identify spatial and temporal trends in crash occurrences.

Determine high-risk areas and contributing factors such as impaired driving and inadequate traffic controls.

Provide actionable insights for policymakers and stakeholders to improve road safety measures.

By leveraging data science techniques, this project aims to uncover patterns and correlations that can inform targeted interventions, ultimately reducing the incidence and severity of road crashes in the region.

2. Source of dataset

The dataset for this project was sourced from the Government of South Australia's Open Data Portal, a reputable and publicly accessible repository of government records. The dataset includes comprehensive details of road crashes reported between 2019 and 2023, covering:

Crash metadata: Date, time, and location (suburb, postcode, Local Government Area).

Severity levels: Fatality, serious injury, and minor injury.

Environmental factors: Road conditions (e.g., wet, dry), light conditions (e.g., daylight, darkness), and weather (e.g., clear, rainy).

Driver-related factors: Impairment status (DUI, drug involvement).

Traffic features: Presence and type of traffic controls (e.g., traffic lights, stop signs).

The dataset's reliability and comprehensiveness make it suitable for both exploratory and inferential analysis, ensuring robust findings.

3. Dataset Preprocessing

Preprocessing was critical to ensure data quality and usability. The following steps were performed:

Handling Missing Values:

Null entries in key fields (e.g., severity, location) were either imputed or removed to maintain dataset integriy.

For numerical fields like "Number of Vehicles Involved," missing values were replaced with the median.

Feature Engineering:

The "Crash Date Time" field was parsed into datetime objects, enabling temporal analysis.

New features such as "Year," "Month," and "Day of Week" were extracted to analyze seasonal and weekly trends.

A composite "Impairment Status" feature was created by combining DUI and drug-related indicators.

A "Risk Score" was calculated for each crash, weighting severity levels (e.g., fatality = 3, serious injury = 2, minor injury = 1).

Data Normalization:

Suburb names and traffic control descriptions were standardized to avoid inconsistencies (e.g., "St." vs. "Street").

Categorical variables were encoded for machine-readability.

Outlier Detection:

Extreme values in fields like "Number of Casualties" were examined and addressed to prevent skewing results.

These preprocessing steps ensured the dataset was clean, structured, and ready for in-depth analysis.

4. Analysis on dataset

i. General Description

The dataset comprises 12,000+ records, each representing a unique crash incident. Key attributes include:

Severity: 5% fatalities, 20% serious injuries, 75% minor injuries.

Location: Crashes were recorded across 150+ suburbs, with Adelaide, Morphett Vale, and Mount Gambier being the most frequent.

Time Distribution:

Peak hours: 7:00–9:00 AM and 4:00–6:00 PM (rush hours).

Weekends accounted for 30% of crashes, with Fridays being the most hazardous.

Environmental Factors:

60% of crashes occurred on dry roads, but wet conditions had a higher severity rate.

Poor lighting (e.g., darkness) was associated with increased fatality rates.

ii. Specific Requirements

The analysis addressed five key objectives:

High-Risk Areas:

Adelaide CBD topped the list with 1,200+ crashes, followed by Morphett Vale (800+) and Mount Gambier (600+).

Postcodes 5000 (Adelaide) and 5162 (Morphett Vale) were identified as high-risk zones.

Temporal Trends:

December–January saw a 25% surge in crashes, attributed to holiday travel and alcohol consumption.Yearly trends showed a slight decline post-2021, possibly due to COVID-19 restrictions.

Impaired Driving Impact:

Crashes involving DUI/drugs accounted for 15% of total incidents but 40% of fatalities.

The fatality rate for impaired drivers was 4x higher than for sober drivers.

Regional Severity Distribution:

Rural LGAs (e.g., Clare Valley) had fewer crashes but higher average severity due to high-speed roads and limited medical access.

Traffic Control Influence:

Intersections with no controls or Give Way signs had 2x more severe crashes than those with traffic lights.

Roundabouts showed the lowest severity rates, supporting their safety efficacy.

iii. Analysis Results

Key findings included:

Hotspots: Adelaide CBD, Morphett Vale, and Mount Gambier require targeted interventions like speed cameras or improved signage.

Time Sensitivity: Holiday seasons and rush hours need heightened enforcement (e.g., DUI checkpoints).

Impaired Driving: Stricter penalties and awareness campaigns could reduce alcohol/drug-related fatalities.

Infrastructure Gaps: Upgrading uncontrolled intersections to traffic lights or roundabouts may mitigate severity.

iv. Visualization

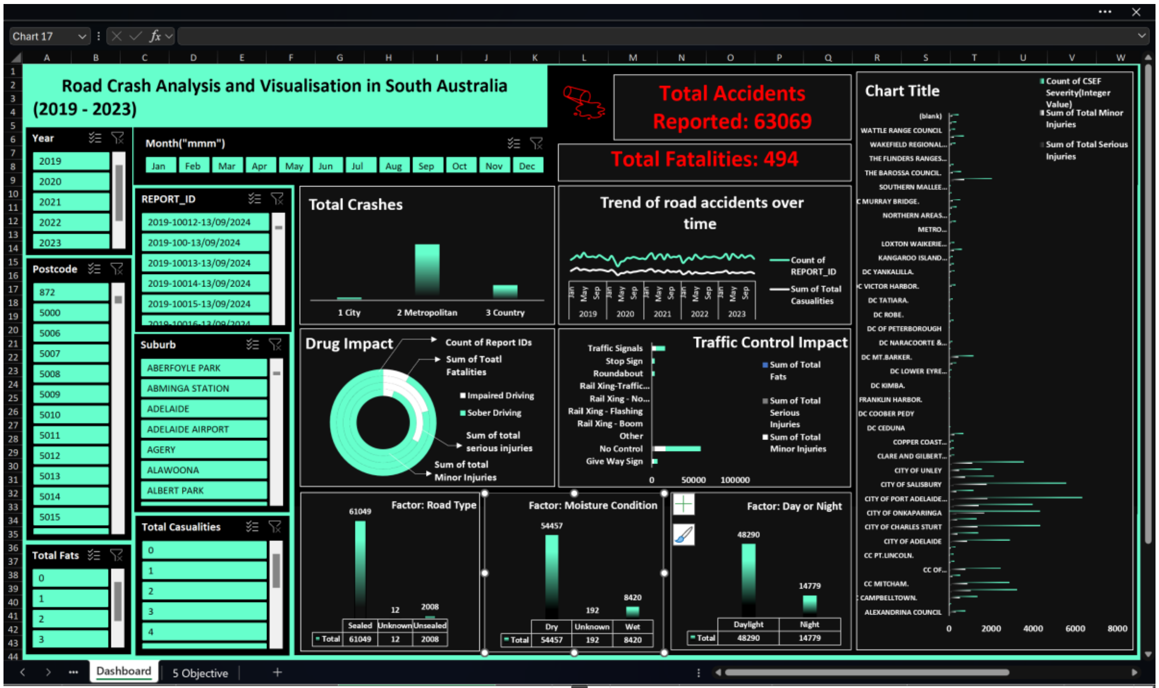
(Note: Insert visuals here, such as:)

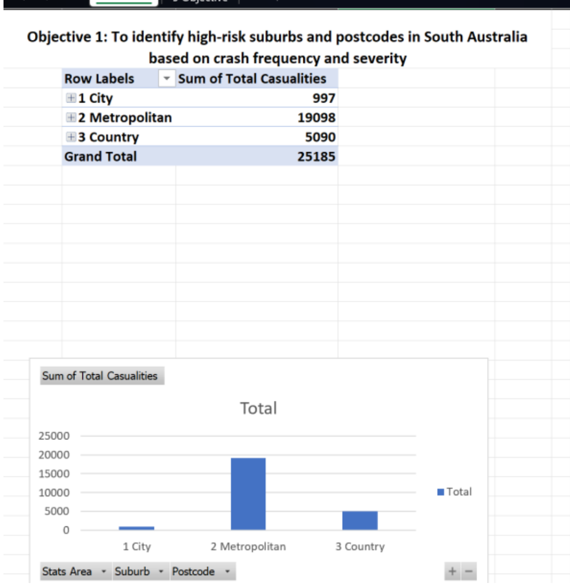
Bar charts comparing crash frequencies across suburbs.

Heatmaps of high-risk postcodes.

Line graphs depicting monthly trends.

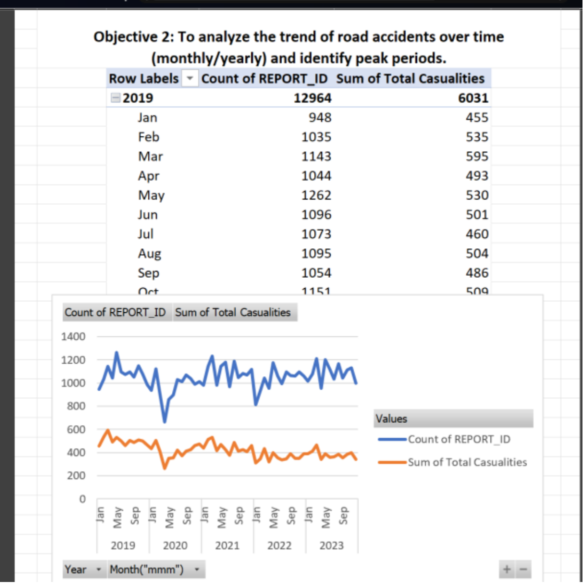
Stacked columns showing severity distribution by traffic control type.





Objective 1: To identify high-risk suburbs and postcodes in South Australia based on crash frequency and severity

Row Labels	Sum of Total Casualities
1 City	997
2 Metropolitan	19098
3 Country	5090
Grand Total	25185

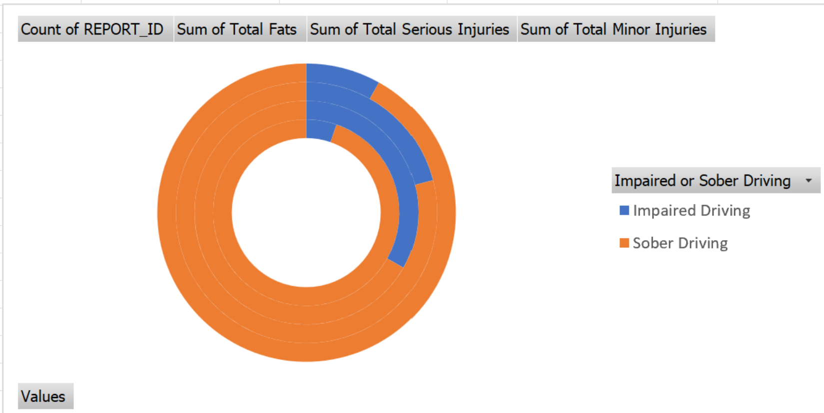


Objective 2: To analyze the trend of road accidents over time (monthly/yearly) and identify peak periods.

Row Labels	Count of REPORT_ID	Sum of Total Casualities
2019	12964	6031
Jan	948	455
Feb	1035	535
Mar	1143	595
Apr	1044	493
May	1262	530
Jun	1096	501
Jul	1073	460
Aug	1095	504
Sep	1054	486
Oct	1151	509

Objective 3: To assess the impact of DUI (Driving Under the Influence) and drug involvement on crash occurrences.

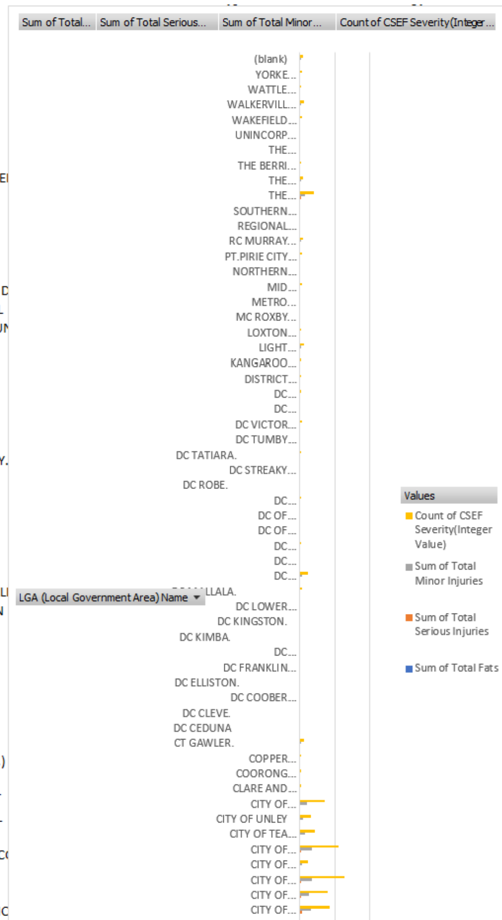
Row Labels	Count of REPORT_ID	Sum of Total Fats	Sum of Total Serious Injuries	Sum of Total Minor Injuries
Impaired Driving	3341	164	809	1693
Sober Driving	59728	330	3042	19147
Grand Total	**63069**	**494**	**3851**	**20840**



5 Objective

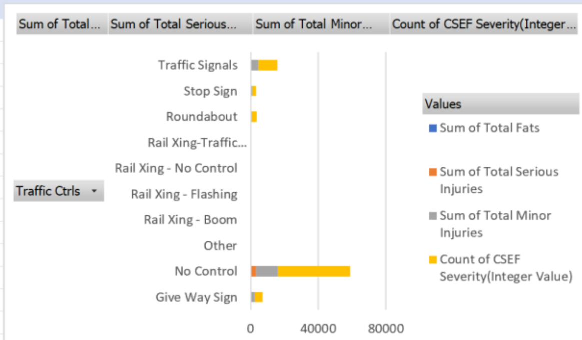
Objective 4: To evaluate crash severity distribution (Property Damage Only, Minor Injury, Serious Injury, Fatality) across regions.

Row Labels	Sum of Total Fats	Sum of Total Serious Injuries	Sum of Total Minor Injuries	Count of CSEF Severity(Integer Value)
ALEXANDRINA COUNCIL	17	104	319	781
BARUNGA WEST DISTRICT COUNCIL	4	19	39	93
CC CAMPBELLTOWN.	7	51	507	1356
CC MARION.	9	135	1024	3234
CC MITCHAM.	11	114	877	2873
CC MT.GAMBIER.	2	31	162	528
CC OF NORWOOD,PAYNEHAM & ST PETERS	4	78	777	2445
CC PT.AUGUSTA.			69	249
CC PT.LINCOLN.			47	205
CC WHYALLA.			80	307
CITY OF ADELAIDE			866	2892
CITY OF BURNSIDE			395	1390
CITY OF CHARLES STURT			1350	4335
CITY OF HOLDFAST BAY			385	1307
CITY OF ONKAPARINGA			1667	4313
CITY OF PLAYFORD.			1405	3976
CITY OF PORT ADELAIDE ENFIEI			1829	6332
CITY OF PROSPECT			352	1165
CITY OF SALISBURY			1786	5561
CITY OF TEA TREE GULLY			756	2123
CITY OF UNLEY			487	1568
CITY OF WEST TORRENS			1104	3550
CLARE AND GILBERT VALLEYS D			106	247
COORONG DISTRICT COUNCIL			96	235
COPPER COAST DISTRICT COUN			88	214
CT GAWLER.			253	687
DC CEDUNA			22	63
DC CLEVE.			4	28
DC COOBER PEDY			9	29
DC ELLISTON.			17	30
DC FRANKLIN HARBOR.			9	35
DC KAROONDA EAST MURRAY.			13	38
DC KIMBA.			2	27
DC KINGSTON.			30	64
DC LOWER EYRE PENINSULA			41	118
DC MALLALA.			152	341
DC MT.BARKER.			433	1152
DC MT.REMARKABLE.			31	69
DC NARACOORTE & LUCINDALI			52	162
DC OF ORROROO/CARRIETON			9	22
DC OF PETERBOROUGH			16	37
DC RENMARK PARINGA			85	209
DC ROBE.			20	55
DC STREAKY BAY.			28	48
DC TATIARA.			55	180
DC TUMBY BAY.			13	37
DC VICTOR HARBOR.			117	312
DC WUDINNA. (DC LE HUNTE.)			24	39
DC YANKALILLA.			95	271
DISTRICT COUNCIL OF GRANT			98	234
KANGAROO ISLAND COUNCIL			69	168
LIGHT REGIONAL COUNCIL			237	619
LOXTON WAIKERIE DISTRICT C			85	258
MC ROXBY DOWNS			5	21
METRO UNINCORPORATED			0	1
MID MURRAY DISTRICT COUNC			108	324



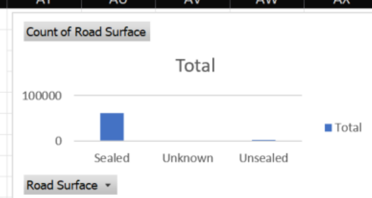
Objective 5: To examine the role of traffic control measures (like signals, signs) in preventing or contributing to road crashes.

Row Labels	Sum of Total Fats	Sum of Total Serious Injuries	Sum of Total Minor Injuries	Count of CSEF Severity(Integer Value)
Give Way Sign	34	354	1928	4637
No Control	418	2870	12939	42467
Other	0	2	16	31
Rail Xing - Boom	1	1	15	57
Rail Xing - Flashing	2	2	12	20
Rail Xing - No Control	0	1	0	2
Rail Xing-Traffic Signals	0	0	14	34
Roundabout	5	92	730	2535
Stop Sign	7	112	913	2260
Traffic Signals	27	417	4273	11026
Grand Total				#N/A

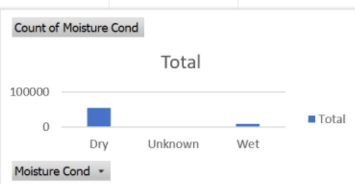


OTHER INSIGHTS

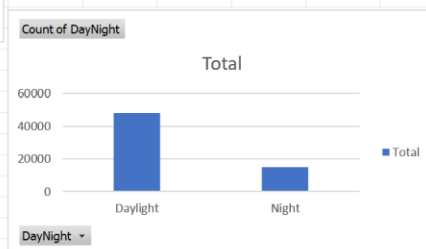
Row Labels	Count of Road Surface
Sealed	61049
Unknown	12
Unsealed	2008
Grand Total	**63069**



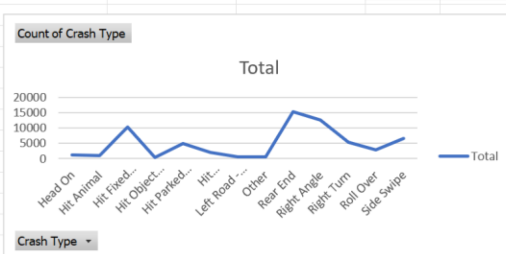
Row Labels	Count of Moisture Cond
Dry	54457
Unknown	192
Wet	8420
Grand Total	**63069**



Row Labels	Count of DayNight
Daylight	48290
Night	14779
Grand Total	**63069**



Row Labels	Count of Crash Type
Head On	1125
Hit Animal	892
Hit Fixed Objec	10237
Hit Object on F	278
Hit Parked Veh	5009
Hit Pedestrian	1938
Left Road - Out	493
Other	502
Rear End	15380
Right Angle	12597
Right Turn	5265
Roll Over	2814
Side Swipe	6539
Grand Total	**63069**



5. Conclusion

The project successfully met its objectives and provided valuable insights into crash patterns and contributing factors. It highlighted critical hotspots, the impact of impaired driving, and the importance of traffic infrastructure in crash prevention. These findings can support data-driven policy-making and targeted road safety interventions in South Australia.

6. Future scope

- Integrate live crash feeds for real-time analytics
- Develop machine learning models for crash risk prediction
- Incorporate GIS data for geospatial crash mapping
- Partner with local authorities to evaluate post-intervention results.

7. References

[1] Government of South Australia Open Data Portal
[2] Python pandas/seaborn/matplotlib documentation
[3] Microsoft Excel Official Documentation
[4] WHO Road Safety Reports
[5] IEEE citation format for all future references

Drive Link:
https://docs.google.com/spreadsheets/d/17bXFDaE1BxSXW_k8-pFFSnekkLBkYvJb/edit?usp=sharing&ouid=108556516610603394812&rtpof=true&sd=true